

St. Michael's University Hospital - Division of Hematology & Oncology
INPATIENT DISCHARGE SUMMARY

PATIENT: COLLINS, Anna

MRN: AML054

DOB: 11-Sep-1952

ADMISSION DATE: 07-Aug-2023

DISCHARGE DATE: 11-Sep-2023

PRIMARY DISCHARGE DIAGNOSIS: Acute Myeloid Leukemia (AML), Philadelphia Chromosome-Positive (Ph+).

SECONDARY DIAGNOSES:

- Induction Chemotherapy-related Neutropenic Fever
- Tumor Lysis Syndrome, Laboratory
- Thrombocytopenia
- Anemia
- Grade 2 Mucositis
- Hypokalemia, Hypomagnesemia

SUMMARY OF HOSPITALIZATION:

Ms. Collins is a 71-year-old female with a recent history of hypothyroidism who presented to the emergency department on August 1, 2023, with complaints of severe fatigue, left upper quadrant abdominal pain, and early satiety. Physical exam was notable for significant splenomegaly. Her initial CBC showed a WBC of 112.5 K/uL with 88% blasts, Hgb 8.1 g/dL, and Platelets 45 K/uL.

She was admitted for an expedited workup. A bone marrow biopsy on August 2, 2023, confirmed a diagnosis of Acute Myeloid Leukemia with 92% blasts.

Key Diagnostic Findings:

- **Flow Cytometry:** Blasts expressed CD34, CD13, CD33, CD117, MPO, and co-expressed CD19 and TdT, consistent with AML with lymphoid markers.
- **Cytogenetics (returned 07-Aug-2023):** Karyotype revealed the presence of the Philadelphia chromosome: **t(9;22)(q34.1;q11.2)** in 20 of 20 metaphases analyzed.
- **FISH:** Interphase FISH was positive for the **BCR-ABL1 fusion gene** in 95% of nuclei.
- **Molecular:** Quantitative RT-PCR for BCR-ABL1 transcript was positive for the p210 transcript. No other high-risk mutations were identified on the NGS panel.
- **Diagnosis:** Philadelphia Chromosome-Positive Acute Myeloid Leukemia. Per ELN 2022 criteria, this is an **Adverse-risk** entity.

Hospital Course:

Following diagnosis, a Hickman catheter was placed. She was started on Allopurinol and aggressive IV hydration for TLS prophylaxis. After extensive discussion of the high-risk nature of her disease and the treatment protocol, she consented to proceed with intensive induction chemotherapy combined with a Tyrosine Kinase Inhibitor (TKI).

Treatment Regimen Initiated 09-Aug-2023:

- **Induction Chemotherapy:** Standard "7+3" regimen (Cytarabine 100 mg/m²/day continuous IV infusion for 7 days and Idarubicin 12 mg/m²/day IV for 3 days).

- **Targeted Therapy: Dasatinib** (a second-generation TKI) 100 mg PO daily, started concurrently on Day 1.

Her course was complex, as expected:

- **Tumor Lysis Syndrome:** Despite prophylaxis, she developed laboratory TLS on Day 2 of induction, with a peak Uric Acid of 10.5 mg/dL and Phosphorus of 5.8 mg/dL. This was managed aggressively with continued hydration and a course of Rasburicase, with normalization of her metabolic panel within 48 hours.
- **Profound Myelosuppression:** She entered her nadir period around Day +8. Her ANC remained < 500 cells/uL for a total of 17 days. Daily CBCs were monitored closely.
- **Neutropenic Fever:** On Day +11, she developed a fever to 39.2°C. Blood cultures were drawn, and she was started on empiric, broad-spectrum IV antibiotics (Meropenem). A chest CT was negative for infiltrate. Blood cultures ultimately remained negative. She defervesced after 4 days and completed a 7-day course of antibiotics.
- **Transfusion Support:** She required significant support through her aplasia:
 - Packed Red Blood Cells: 11 units total to maintain Hgb > 7.0 g/dL.
 - Apheresis Platelets: 18 units total to maintain Plt > 10 K/uL.
- **TKI Side Effects:** She developed a mild pleural effusion around Day +15, a known side effect of Dasatinib. This was managed conservatively by temporarily holding the drug for 5 days and a short course of diuretics, with complete resolution seen on chest X-ray. Dasatinib was successfully resumed at the same dose.

Response and Recovery:

Peripheral blood counts began to recover around Day +28. A Day +35 bone marrow biopsy was performed (04-Sep-2023) to assess response.

- **Pathology Report (07-Sep-2023):**
 - **Morphology:** < 5% blasts, trilineage hematopoiesis present. Morphologic Complete Remission.
 - **FISH:** BCR-ABL1 fusion signal was detected in only 1% of nuclei (a major cytogenetic response).
 - **Quantitative PCR:** BCR-ABL1 transcript level had fallen dramatically by >3 logs.

She became transfusion-independent on September 6th. Her ANC is now robust. She is ambulatory, tolerating a diet, and eager to go home.

Condition at Discharge:

Clinically well. Afebrile, hemodynamically stable. Hickman line is clean, patent, and dressing is intact.

- **Discharge Labs (11-Sep-2023):** WBC 5.6 K/uL, ANC 3.1 K/uL, Hgb 9.8 g/dL, Plt 121 K/uL.

Discharge Plan:

1. **Medications:**
 - **Dasatinib 100 mg** PO daily. Patient educated on importance of adherence.
 - **Levofloxacin 500 mg** PO daily for antibacterial prophylaxis.
 - **Voriconazole 200 mg** PO BID for antifungal prophylaxis.
 - **Acyclovir 400 mg** PO BID for antiviral prophylaxis.

- **Pantoprazole 40 mg** PO daily.
- 2. **Follow-Up:**
 - Appointment in our Leukemia Clinic in 1 week (18-Sep-2023) for lab check and clinical assessment and planning of allogeneic SCT (MUD 10/10 donor available)
- 3. **Line Care:** Visiting Nurse Association (VNA) referral placed for twice-weekly Hickman dressing changes and flushes.

The patient and her family received extensive education regarding her medications, line care, and strict precautions for infection prevention. Prognosis remains guarded due to the high-risk nature of Ph+ AML, but her initial response to therapy is exceptionally encouraging.

Dr. Benjamin Carter, MD
Hematology-Oncology Service